



# HAYK SAHAKYAN

Data Scientist / AI Engineer

## Experience

29.08.1991

+374 94 808 802

hayk.ashoti.sahakyan@gmail.com

[linkedin.com/in/hayk-sahakyan-021471138](https://www.linkedin.com/in/hayk-sahakyan-021471138)

## Language

- Armenian
- English
- Russian

### 2025 - Fast Bank CJSC Data Scientist Expert

- **Scorecard Development & Validation:** Designing and validating credit scorecards using statistical and ML methods.
- **Risk Analytics:** Analyzing loan portfolios, default behavior, and customer segments to support credit risk decisions.
- **Data Engineering for Risk Modeling:** Automating data pipelines, integrating multiple data sources.
- **Financial Risk Framework:** Building a Streamlit-based internal dashboard for scoring, monitoring, and reporting.

### 2022-2024 Cognize CJSC Data Scientist

- **Natural Language Processing (NLP):** Developed an NLP model using RNNs, LSTMs, and bidirectional networks to tokenize and predict row structures in tabular data.
- **Table Structure Detection:** Designed a rule-based model to detect table structures in OCR-processed documents using word/symbol coordinates and projection techniques.
- **Financial Reports Classification:** Implemented a multilingual BERT-based model to classify financial report sections like income statements and balance sheets, improving document organization and retrieval.
- **Form Data Extraction:** Extracted data from mixed handwritten and typed German forms using OCR techniques, GPT-4 APIs, and statistical analysis, achieving high data accuracy.
- **Tax Reports Classification:** Utilized ResNet and domain insights to classify tax documents effectively.
- **Automation:** Developed internal tools using Streamlit and Flask for database queries, data uploads, and API calls, streamlining operations.
- **Financial Spreading:** Automated financial statement population (income, balance sheet, cash flow) using BERT and LLAMA.

### 2021-2022 Cognize CJSC Python Developer

- Conducted web scraping projects to gather legal and financial data using Selenium and BeautifulSoup, ensuring data accuracy and relevance.
- Processed and structured scraped data into MongoDB, creating actionable datasets for client-defined tasks.
- Built pipelines for automated document generation from processed data.

### 2020 - 2021 Cognize CJSC - Financial Analyst

- Reviewed AI-processed financial documents to ensure data accuracy and compliance.
- Contributed to data quality improvement initiatives for financial reports.

### 2017 - 2020 SCDM LLC (Deloitte Audit Analytics GmbH Subsidiary) - Financial Analyst

- Modeled structured finance products, including bonds and derivatives, for data processing and analysis.
- Processed and validated complex financial datasets to support structured finance teams.

## Skills

#### Programming and Frameworks

- **Programming Languages:** Python (NumPy, Pandas), R
- **ML & DL Frameworks:** PyTorch, TensorFlow, Scikit-learn
- **Web Development:** Flask, Streamlit, HTML5/XML

#### Data Engineering & Tools

- **Data Manipulation and Web Scraping:** Selenium, BeautifulSoup
- **Big Data Technologies:** PySpark
- **Databases:** SQLite, MySQL, MongoDB

#### Data Visualization

- Plotly, Matplotlib

#### Advanced Topics

- Large Language Models (Transformers, BERT, LLaMA), Generative AI (VAE, Diffusion Models), Computer Vision (CNNs, ResNet, ViT, DINO), **Graph Neural Networks**, Natural Language Processing (**NLP**), Digital Signal Processing (**DSP**),

#### Version Control and Operating Systems

- Git/GitHub, Linux

## Education

- 2023 - 2025

Yerevan State University, Department of *Mathematics and Mechanics*, **Applied Statistics and Data Science Master's** degree

- 2016-2018

Armenian State University of Economics, Financial Department, *Corporate Finances*, **Master's** degree

- 2013-2016

Armenian State University of Economics, Financial Department, Finance, *Bachelor's* degree

- 2008-2010

Armenian State University of Economics, Informational and Statistical Department, Informational system management unit

## DIPLOMAS, CERTIFICATES, TRAININGS

- Armenia LLM Summer School 2026 (ArmLLM), Yerevan — hosted at AI9 Startup Campus (2026 Aug). Intensive week on LLM pre-training and scaling laws, Mixture-of-Experts, supervised fine-tuning and reinforcement learning, test-time scaling and compute-optimality, agents and reasoning workflows, and vision-language-action models.
- "AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents" — Udemy (2026). Hands-on program building 8 end-to-end LLM applications: frontier and open-source model integration, RAG pipelines with vector databases, QLoRA fine-tuning of open-source models, agentic multi-model workflows, and deployment with Gradio.
- Certificate for the successful completion of **Fundamentals of Deep Learning** awarded by **NVIDIA Deep Learning Institute** (2022 Jun)
- Certificate for completing 5 months duration **"Machine Learning Intermediate"** course offered by the Ministry of High Tech Industry of the Republic of Armenia and Armenian Code Academy (ACA) (2021 Oct – 2022 Mar)
- Certificate of Excellence for completing 106 hours duration beginner level **"Machine Learning"** course offered by the Ministry of High Tech Industry of the Republic of Armenia and Armenian Code Academy (ACA) (2020 Dec – 2021 Mar)
- Certificate for successful completion of **"Advanced Python"** 12-week course offered by Armenian Code Academy (ACA) (2020 Mar – 2020 Jul)
- Certificate for successful completion of **"Introduction to Python"** 8-week course offered by Armenian Code Academy (ACA) (2019 Nov – 2020 Jan)